

1. Administrative & Study Identification

Full Study Title: A Phase 1b/2 Dose Escalation and Expansion Study of Tucatinib in Combination With Trastuzumab and Oxaliplatin-based Chemotherapy or Pembrolizumab-containing Combinations for HER2+ Gastrointestinal Cancers **Short Title/ Acronym:** Tucatinib Plus Trastuzumab and Chemotherapy/ Pembrolizumab for HER2+ GI Cancers **Protocol Number:** SGNTUC-024 **NCT Number:** NCT04430738 **Sponsor Name:** Pfizer **Company:** Pfizer **Investigational Product:** Tucatinib (TUKYSA), Trastuzumab, Oxaliplatin-based Chemotherapy (FOLFOX/CAPOX), Pembrolizumab **Phase of Development:** PHASE2 (Phase 1b/2) **Trial Status:** ACTIVE_NOT_RECRUITING (Active but not currently recruiting) **Medical Monitor:** Pfizer Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate the safety, tolerability, and specific toxicities (including renal dose-limiting toxicities and treatment-emergent adverse events) of tucatinib in combination with trastuzumab and oxaliplatin-based chemotherapy (FOLFOX or CAPOX), or pembrolizumab. **Secondary Objectives:** To evaluate preliminary anti-tumor activity (Objective Response Rate [ORR] per RECIST v1.1, Progression-Free Survival [PFS], Overall Survival [OS], Duration of Response [DoR]), to assess the pharmacokinetics of the combinations, and to monitor the incidence of TEAEs across different cohorts.

2.2 Background & Rationale Human epidermal growth factor receptor 2 (HER2) gene amplification or protein overexpression (HER2+) occurs in a significant subset of gastrointestinal (GI) cancers, including various forms of adenocarcinoma. While HER2-targeted therapies like trastuzumab have improved survival, mechanisms of resistance eventually cause disease progression. Tucatinib is a highly selective HER2-directed tyrosine kinase inhibitor. This study explores combining this dual blockade with standard-of-care oxaliplatin-based chemotherapy or the immune checkpoint inhibitor pembrolizumab to further enhance therapeutic efficacy in patients with unresectable or metastatic HER2+ GI cancers.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Single-arm (Multiple Dose-escalation and Expansion Cohorts) **Blinding:** Open-label **Randomization:** N/A

3.2 Planned Enrollment & Duration Targeted Enrollment: 40 participants
Number of Sites: 27 locations globally
Estimated Study Start: Mid-2020
Current Trial Status: ACTIVE_NOT_RECRUITING (Active but not currently recruiting)

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Unresectable or metastatic solid malignancy that is histologically or cytologically confirmed to be one of the tumor types listed below: * **Cohorts 1A, 1B, 1C, and 1D:** Colorectal Carcinoma (CRC), Gastric Adenocarcinoma, GEJ Adenocarcinoma, Esophageal Adenocarcinoma, Cholangiocarcinoma, Gallbladder Carcinoma. * **Cohorts 1E, 1F, 1G, and 2A:** Gastric Adenocarcinoma, GEJ Adenocarcinoma, Esophageal Adenocarcinoma. * **Cohort 2B:** Colorectal Carcinoma (CRC). 2. Must be a candidate to receive an oxaliplatin-based regimen as part of their standard-of-care treatment for all cohorts, except Cohort 1G. 3. HER2+ disease, as determined by historic or local laboratory testing. 4. **Phase 1b cohorts:** Measurable or non-measurable disease according to RECIST v1.1 as determined by the investigator. 5. **Phase 2 cohorts:** Measurable disease according to RECIST v1.1 as determined by the investigator. 6. Eastern Cooperative Oncology Group (ECOG) Performance Status score of 0 or 1.

4.2 Exclusion Criteria 1. History of known hypersensitivity to the planned study treatments. 2. Known to be positive for Hepatitis B or C infection. 3. For Cohorts 2A and 2B: Prior anti-HER2 therapies. 4. For Cohorts 1E, 1F, 1G, 2A: Prior therapy with an anti-PD-1, anti-PD-L1, or anti-PD-L2 agent or with an agent directed to another stimulatory or co-inhibitory T-cell receptor (eg, CTLA-4, OX 40, CD137), that was discontinued due to a Grade ≥ 3 immune-related adverse event (irAE).

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: * **Tucatinib:** Administered orally. * **Trastuzumab:** Administered via intravenous (IV) infusion. * **Chemotherapy/Immunotherapy Backbones:** FOLFOX (includes fluorouracil [trade names: Carac, Efudex, Adrucil, Fluoroplex, Tolak], leucovorin, oxaliplatin), CAPOX, or Pembrolizumab. **Route of Administration:** Oral (Tucatinib, Capecitabine) and IV (Trastuzumab, Oxaliplatin, Fluorouracil, Leucovorin, Pembrolizumab). **Duration of Treatment:** Treatment continues until unacceptable toxicity, documented disease progression, investigator decision, or participant withdrawal of consent.

5.2 Concomitant & Prohibited Therapy Permitted: Standard antiemetic treatments and antidiarrheal prophylaxis. **Prohibited:** Use of other concurrent systemic anticancer therapies not specified in the protocol.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -28 to 0	Informed Consent, Baseline Imaging, HER2+ confirmation, Medical History, Labs
Baseline	Cycle 1, Day 1	First Dose of Investigational Regimen, Vital Signs, ECG
Interim	Every 14 or 21 Days	Cycle-based dosing, Safety Labs (including Serum Cystatin C), PK/PD Sampling
Tumor Assessment	Every 6 Weeks	Imaging for Efficacy Assessment (RECIST v1.1)
End of Study	+30 Days post-dose	Final Vital Signs, Exit Interview, IP Accountability, 30-Day Safety Follow-up

7. Outcome Measures (Endpoints)

7.1 Primary Endpoints

- 1. Number of Participants With Renal Dose-Limiting Toxicities (DLTs): Phase 1b (Cohort 1A and 1B).** A renal DLT is defined as an increase in serum cystatin C >1.5 baseline not related to pre-renal or post-renal etiologies occurring between the first dose of tucatinib and the end of Cycle 3.
- 2. Number of Participants With Treatment Emergent Adverse Events (TEAEs): Phase 1b (Cohort 1E and 1F).** Defined as newly occurring or worsening AEs after the first dose of study treatment through 30 days after the last dose.
- 3. Number of Participants With Treatment Related TEAEs: Phase 1b (Cohort 1E and 1F).*** TEAEs judged by the investigator to be related to any study treatment.

7.2 Secondary Endpoints

- 1. Number of Participants With TEAEs: Phase 1b (Cohort 1A and 1B).**
- 2. Number of Participants With Treatment Related TEAEs: Phase 1b (Cohort 1A and 1B).**
- 3. Number of Participants With $\geq$ Grade 3 TEAEs: Phase 1b (Cohort 1A and 1B).**
- 4. Efficacy Outcomes:** Objective Response Rate (ORR), Progression-Free Survival (PFS), Overall Survival (OS), and Duration of Response (DoR).
- 5. Pharmacokinetics (PK):** Exposure profiles of tucatinib, trastuzumab, and backbone agents.

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Continuous monitoring from the first dose of study treatment through 30 days after the last dose. **Renal Toxicity Focus:** Strict monitoring of serum cystatin C for potential renal DLTs, distinguishing treatment-related effects from intercurrent illness or disease progression. **Data Monitoring Committee (DMC):** A Sponsor-led Safety Review Committee (SRC) evaluates sequential dose-escalation safety data to make determinations on cohort expansion and RP2D declarations.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: The Safety Set includes all enrolled patients who receive at least one dose of any study drug. The Efficacy Evaluable population includes patients who receive study drug and have baseline and at least one post-baseline tumor assessment. **Sample Size Justification:** A standard dose-escalation statistical design with subsequent expansion cohorts (Targeted Enrollment = 40). **Handling of Missing Data:** Time-to-event efficacy variables analyzed using Kaplan-Meier methodology.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: Conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Risk-based monitoring with a mix of onsite and remote visits to ensure protocol adherence. **Audit Trail:** Robust Electronic Data Capture (EDC) with eCRF locking and Source Data Verification.

11. Study Identifiers & Governance

Posting ID: 276836765530532585 **Primary ID:** SGNTUC-024 **Registry Number:** NCT04430738 **Sponsor/Company:** Pfizer **Study Title:** Tucatinib Plus Trastuzumab and Oxaliplatin-based Chemotherapy or Pembrolizumab-containing Combinations for HER2+ Gastrointestinal Cancers **Official Regulatory Title:** A Phase 1b/2 Dose Escalation and Expansion Study of Tucatinib in Combination With Trastuzumab and Oxaliplatin-based Chemotherapy or Pembrolizumab-containing Combinations for HER2+ Gastrointestinal Cancers **Trial Status:** ACTIVE_NOT_RECRUITING

12. Clinical Framework (Oncology Specific)

Disease Areas: Colorectal Carcinoma, Gastric Adenocarcinoma, GEJ Adenocarcinoma, Esophageal Adenocarcinoma, Cholangiocarcinoma, Gallbladder Carcinoma. **Preferred UMLS Name:** HER2 Positive Gastrointestinal Neoplasms **Morphology/Disease Name:** Adenocarcinoma **Diagnosis Threshold:** HER2 gene amplification or protein overexpression (HER2+) via historic or local laboratory testing. **Medical Methodology:** Dual HER2-targeted therapy combined with Cytotoxic Chemotherapy or Immune Checkpoint Inhibition.

13. Study Procedures & Methodology

Management Pathway: Standard clinical pathway for advanced/metastatic HER2+ solid tumors. **Education Protocol (HCP):** Site initiation visits covering protocol-specific dosing, irAE/GI/renal toxicity management, and reporting criteria. **Education**

Protocol (Patient): Informed consent process, oral medication dosing diaries, and AE reporting guidelines. **Quality Audit Mechanism:** Central/local RECIST v1.1 imaging review, Source Data Verification, and scheduled SRC dose-escalation reviews.

14. Observation & Follow-Up Schedule

Total Duration: Continuous treatment until disease progression or unacceptable toxicity, plus 30-day safety follow-up and long-term survival follow-up. **Onsite Visit Cadence:** Day 1 of each cycle, depending on specific cohort backbone. **Checklist Review Interval:** Tumor imaging assessments scheduled every 6 weeks.

15. Sign-off & Approval

Role	Name	Signature	Date		:---	:---	:---	:---
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]					
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]					
Statistician	[Name]	_____	[DD-MMM-YYYY]					